LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Servo altimeter and cabin altimeter
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Servo altimeter and cabin altimeter.
5. Specific Objectives	15.1.4.1 Trainee will be able to explain Servo altimeter, cabin altimeter.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument schematics; Safety: Do not touch live components; Supervision: Instructor monitors PPT navigation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an altimeter?
(b) How does static pressure change with altitude?
(c) Why is a servo mechanism used in some altimeters?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Servo altimeter operating principle • Servo motor function and feedback loop • Cabin pressure control mechanism • Servo valve operation details • Difference between direct and servo types	L & Ds	PPT, IB	15 min
(b)	• Cabin altimeter installation location • Cabin pressure regulation systems • Comparison of servo and cabin types • Maintenance requirements for servo units • Troubleshooting common servo faults	L & Ds	PPT, PDF	20 min
(c)	• Impact of temperature on readings • Calibration procedures for cabin altimeters • Safety implications of pressure failure • Integration with other flight instruments • Future trends in altimeter technology	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the servo motor in maintaining cabin pressure.
(b) Describe the key difference between a direct reading and a servo altimeter.

12. Home Assignment and Follow Up	Prepare a brief note on the dismantling and assembling procedure of typical capsule instruments as covered in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Dismantling and assembling procedure of typical capsule instruments.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Engineer, P No. 12345	Senior Instructor, Chief Engineer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min